

CSPC Gate Pass & Vehicle Registration System

System Specification (Form CSPC-F-GSU-18)

Project: Smart Campus Vehicle Management System (VMS)
Institution: Camarines Sur Polytechnic Colleges (CSPC)
Document Version: 1.1 - includes GSU office visit step for approved users
Date: August 2026

1. Purpose & Scope

This specification defines the campus vehicle registration and gate access system aligned with **CSPC Form CSPC-F-GSU-18**. The system enables:

- Temporary gate access for unregistered RFID/ID taps at the entrance
- Online vehicle registration with document verification
- Admin review, approval with Control Number assignment, and GSU office finalization
- Resubmission workflow when applications are denied

Technology stack (implementation target): Laravel 12 web portal, MongoDB, ESP32 RFID gates, optional AI parking integration.

2. Actors

Actor	Description
Unregistered ID Holder	Person who taps an unregistered campus ID/RFID at the gate
Applicant (Student/Staff)	User completing CSPC-F-GSU-18 online registration
Admin (GSU/Security)	Reviews applications, approves/denies, assigns Control No.
Guard	Monitors gate, assists visitors, manual overrides
ESP32 Gate Hardware	Reads RFID at entry/exit, communicates with Laravel API
Email System (SMTP)	Sends verification links, approval/decline notifications

3. Functional Requirements

3.1 Unregistered ID Tap - Gate Entrance Flow

Trigger: An unregistered ID card or RFID is tapped at the **entrance gate**.

Step	System Behavior
1	Detect unregistered UID via `POST /api/rfid/scan`
2	Issue a Temporary Gate Pass valid for a configurable duration (default: 2 hours)
3	Display or print a QR code / registration link directing the user to the web portal
4	Grant one-time or time-limited entry while the pass is active
5	If registration is not completed before expiry -> invalidate pass and block future

Business rules:

- Temporary pass duration is **dynamic** (admin-configurable in System Settings).

- Blocked UIDs receive `access\_denied` with reason: **"Temporary pass expired - complete vehicle registration."**
- QR/link target: `/register` (or deep-link with pass token for tracking).

Data entities:

- `temporary\_gate\_passes`: `rfid\_uid`, `issued\_at`, `expires\_at`, `registration\_completed`, `blocked`

3.2 User Web Registration & Email Verification**3.2.1 Account Setup**

Requirement	Detail
Credentials	Email address + user-created password (no SMS)
Verification	System sends OTP or email verification link to the provided address
Post-verification status	Account saved as Email Verified / Registration Pending Admin Review

Status model:**3.2.2 Login & Pending Dashboard (On Denial)**

When admin **denies/rejects** a registration:

Rule	Behavior
Account remains active	User retains Email Verified status
Login allowed	User may sign in with registered email + password
Dashboard state	Pending / Resubmission Dashboard showing: Mandatory Rejection Reason (submitted by admin) Editable form fields for flagged items (Driver's License, vehicle info, uploaded
Resubmission	On resubmit -> status returns to Pending Admin Verification

Important: Denial does **not** lock the account or force a new registration after a cooldown period.

3.3 License Scanner - GCash-Style OCR Integration

Feature	Specification
UI	Active camera viewfinder inside the registration form (similar to GCash KYC/ID s
Capture target	Physical Driver's License or valid ID
OCR auto-fill fields	Applicant Full Name, Address, Contact Number, Driver's License Number
Fallback	Manual entry if OCR confidence is low; user may retake photo

Technical notes:

- Browser `getUserMedia` for live camera preview
- Server-side OCR (EasyOCR/Tesseract) or client-side capture + upload
- Separate from existing Campus ID OCR (which reads SN/name from school ID)

3.4 Web Registration Form - CSPC-F-GSU-18 Fields

The online form collects all parameters matching **CSPC Form CSPC-F-GSU-18**:

Applicant Details

- Full Name
- Address
- Contact Number
- Date of Application (auto-filled: submission date)
- Email Address
- Password

Vehicle Details

- Type of Vehicle
- Model
- Plate Number

Required Document Uploads

1. Clear photo of **Valid Identification / School ID**
2. Clear photo of **Driver's License**
3. Clear photo of **LTO Official Receipt (OR)**
4. Clear photo of **LTO Certificate of Registration (CR)**

Validation: All four uploads required before submission. Image formats: JPEG/PNG/PDF; max size per system settings.

3.5 Admin Review & Verification Workflow

3.5.1 Approval Dashboard

Admin reviews submitted user details and uploaded proof documents.

If Approved:

Step	Action
1	Admin enters assigned Control No. into the system
2	System marks account status as Approved
3	System sends approval email instructing user to visit the General Services Unit
	- Finalize payment
	- Process vehicle sticker
	- Activate full RFID gate access for their tapped ID
4	`Gate_access` remains Pending until GSU confirms payment/sticker (or admin n

Approval email must include:

- Control Number
- GSU office location/hours
- List of documents to bring (if any)
- Reminder that gate access activates after GSU processing

3.5.2 Denial Workflow

If Denied:

Step	Action
1	System forces admin to enter a Mandatory Reason for Rejection (e.g., Driver's License)
2	Application status -> Pending Corrections
3	Rejection reason stored on user record and displayed on resubmission dashboard
4	Decline email sent with rejection reason and link to resubmit

Admin rejection reason field: Required, min 10 characters, max 500 characters.

4. User Status & Access Matrix

Status	Email Verified	Can Login	Dashboard	Gate Access
Email Unverified	No	Limited (verify only)	Verify email page	Denied
Pending Admin Review	Yes	Yes	Waiting for review	Denied (or temp pass only)
Pending Corrections	Yes	Yes	Resubmission form + rejection reason	Denied
Approved (pre-GSU)	Yes	Yes	Approved - visit GSU	Pending
Approved (post-GSU)	Yes	Yes	Full user dashboard	Granted
Temp Pass Active	N/A	N/A	N/A	Time-limited Granted
Temp Pass Expired (unregistered)	N/A	N/A	N/A	Blocked

5. Process Flows

5.1 Unregistered Tap -> Registration -> GSU Activation

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5.2 Denial & Resubmission Flow

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6. API & Integration Points

Endpoint	Purpose
`POST /api/rfid/scan`	Gate tap; issue temp pass if unregistered
`POST /register`	Submit CSPC-F-GSU-18 registration
`POST /register/scan-license`	DL OCR auto-fill (new)
`GET /email/verify/{id}/{hash}`	Email verification link

`POST /admin/registrations/{id}/approve`	Approve + Control No.
`POST /admin/registrations/{id}/decline`	Deny + mandatory reason
`GET /user/resubmit`	Resubmission dashboard (new)
`POST /user/resubmit`	Resubmit corrected application (new)

7. Data Model Additions

Collection / Field	Purpose
`users.control_number`	GSU-assigned Control No. on approval
`users.rejection_reason`	Mandatory admin reason on denial
`users.registration_status`	Extended: `pending_review`, `pending_corrections`, `approved_pre_gsu`, `approved`
`users.address`	Applicant address (CSPC-F-GSU-18)
`users.or_photo`, `users.cr_photo`	Separate OR and CR uploads
`temporary_gate_passes`	Temp pass records for unregistered taps
`system_settings.temp_pass_duration_hours`	Configurable pass duration (default: 2)

8. Email Templates

Template	Trigger	Key Content
Email Verification	Registration	Verification link or OTP
Registration Approved	Admin approve	Control No., **visit GSU office** for payment/sticker
Registration Denied	Admin deny	Mandatory rejection reason, resubmit link
Temp Pass Issued	Unregistered tap	QR/link to register, expiry time

9. Non-Functional Requirements

- ****Security:**** CSRF on forms; token auth on RFID API; hashed passwords; private document storage
- ****Availability:**** Gate must respond within 2 seconds for RFID scan
- ****Usability:**** Mobile-friendly registration; camera OCR works on modern browsers
- ****Audit:**** All gate taps and registration status changes logged

10. Implementation Gap Analysis (Current vs Spec)

Requirement	Current System	Gap
Temp gate pass on unregistered tap	Denies with "Card Not Registered"	**Build temp pass module**
Email + password registration	Implemented	-
Email verification (link)	Implemented (Laravel MustVerifyEmail)	OTP optional
Composite status label	Split: `email_verified_at` + `status`	Add unified status display
Denial -> resubmission dashboard	Denied users blocked; 3-day cooldown	**Replace with resubmission flow**
Mandatory rejection reason	Optional `remarks`	**Make required; persist on user**
DL camera OCR (GCash-style)	Campus ID OCR only; DL is upload	**Add DL scanner + OCR**

Separate OR/CR uploads	Single `or_cr_photo` field	**Split into two fields**
Address field	Missing	**Add to form + model**
Control No. on approval	Not captured	**Add admin input + storage**
GSU office visit email	Generic approval email	**Update template**
Pending Corrections status	Uses `Denied`	**Add new status value**

11. Acceptance Criteria

1. Unregistered RFID tap at gate receives 2-hour temp pass and registration QR within 3 seconds.
2. User can register with email/password, verify email, and reach Pending Admin Review state.
3. Admin cannot approve without entering Control No.; approval email mentions GSU office visit.
4. Admin cannot deny without entering rejection reason (min 10 chars).
5. Denied user can log in, see rejection reason, edit flagged fields, and resubmit.
6. Registration form matches CSPC-F-GSU-18 fields with four separate document uploads.
7. DL camera scanner auto-fills name, address, contact, and license number when readable.
8. Expired temp pass without registration blocks future gate taps from that UID.

End of Specification